

TOI-1516b

R-2048

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1516_241206 · created 2026-08-08T23:54:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460650.712 +/- 0.039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.048
Transit depth (Rp/Rs)^2	2.46 +/- 1.51 [%]
Orbital Inclination (inc)	86.47 +/- 1.44
Airmass coefficient 1 (a1)	1.071 +/- 0.023
Airmass coefficient 2 (a2)	-0.04 +/- 0.092
Scatter in the residuals of the lightcurve fit is	2.11 %
Best Comparison Star	#2 - [438, 353]
Optimal Aperture	3.01
Optimal Annulus	4.39
Transit Duration (day)	0.084 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.048 vs 0.1224 (deviation 0.72σ)
Duration fit vs published	0.084 d vs 0.1217 d (deviation 1.51σ)
Mid-time O–C	31.3 min
Depth SNR	3.3
Residual scatter	2.11 %

Light curve

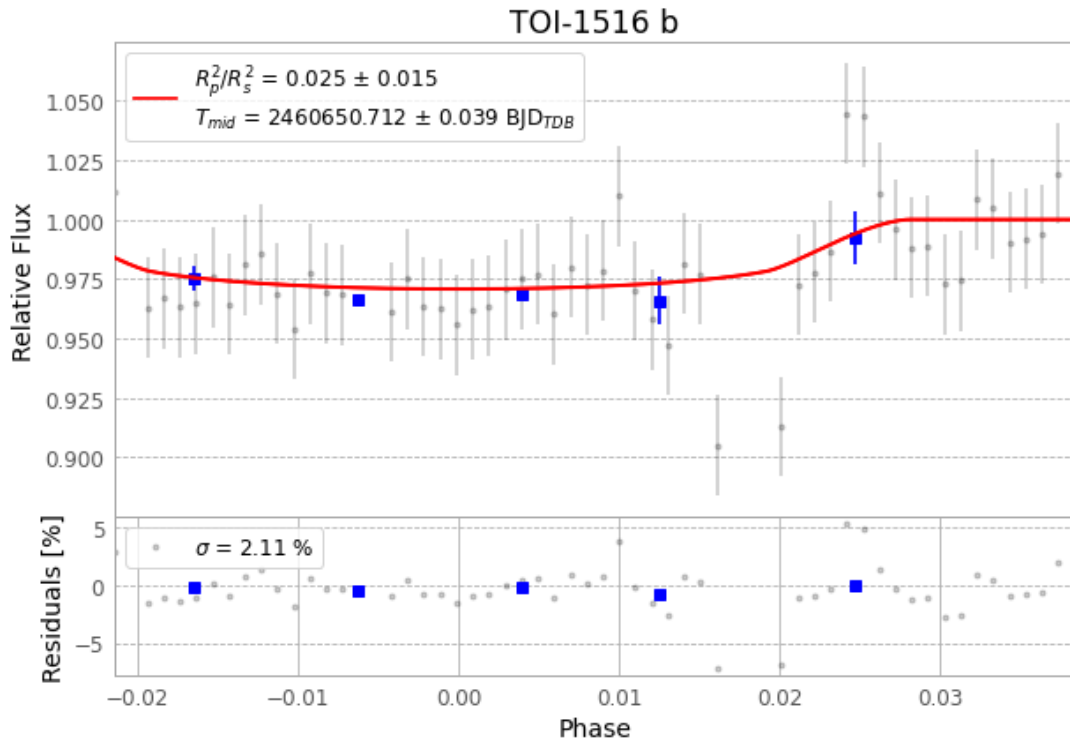


Figure 1 — FinalLightCurve_TOI-1516_b_2024-12-05.png (SHA-256 77dd53f6bdc93a86...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [358, 295], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 898cdba67150e8b5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

60 FITS frames (39 MB), TOI1516241206035715.FITS → TOI1516241206065417.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1516-241206.log (63bac587b284...), FinalLightCurve_TOI-1516_b_2024-12-05.png (77dd53f6bdc9...), FinalLightCurve_TOI-1516_b_2024-12-05.csv (5d75a7331a93...)

Compute resources

Wall time 347.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 23:54Z.